

PH101: 2023

Special Theory of Relativity

Lecture-1

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Syllabus, References, Plan

Special Theory of Relativity:

Postulates of STR, Galilean transformation. Lorentz transformation. Simultaneity. Length Contraction. Time dilation. Relativistic addition of velocities. Energy momentum relationships.

Text / References:

1. D. Kleppner and R. J. Kolenkow, *An Introduction to Mechanics*, Tata McGraw-Hill, 2000.
2. R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics, Vol. I*, Norosa Publishing House, 1998.
3. R. Resnick, *Introduction to Special Relativity*, John Wiley, Singapore, 2000.

There will be five lectures and two and half tutorials to cover them.

A common example of relativity !



1:00



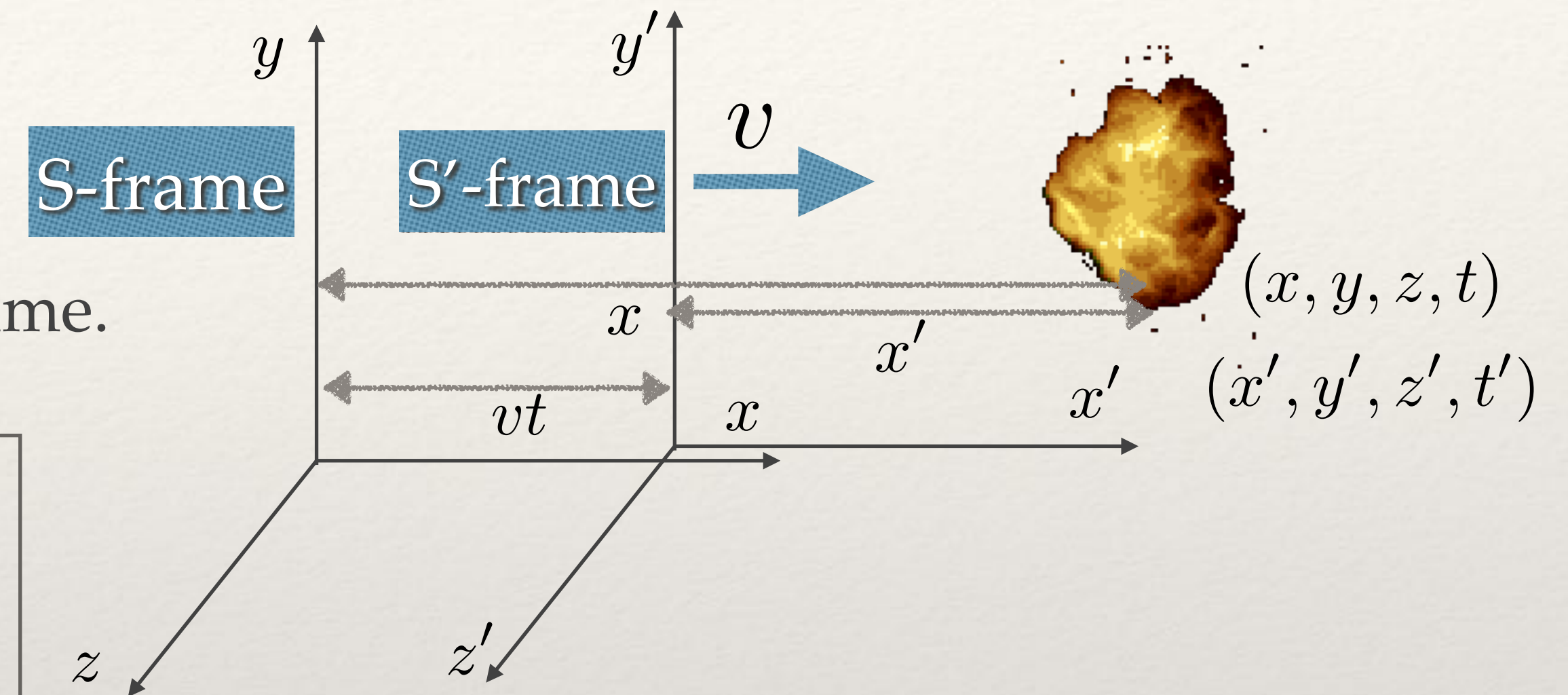
“When a man sits with a pretty girl for an hour, it seems like a minute. But let him sit on a hot stove for a minute — and it's longer than any hour. That's relativity.”: Einstein

Events and reference frames

An event is a phenomenon that happens independent to the state of the observer. Example, an explosion.

Event happens in a position in space and at an instant in time.

Therefore we need four coordinates: three spatial coordinates and one time coordinate of a reference frame to measure it.



Now consider another frame moving with velocity v along x-axis.

Question: How do the coordinates relate to each other ?

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t$$

Galilean transformations

All the reference frames which are at rest or moves with an uniform velocity, where Newton's law of inertia holds are called **inertial frames: non-accelerating frames**

$$t_A - t_B = t'_A - t'_B$$

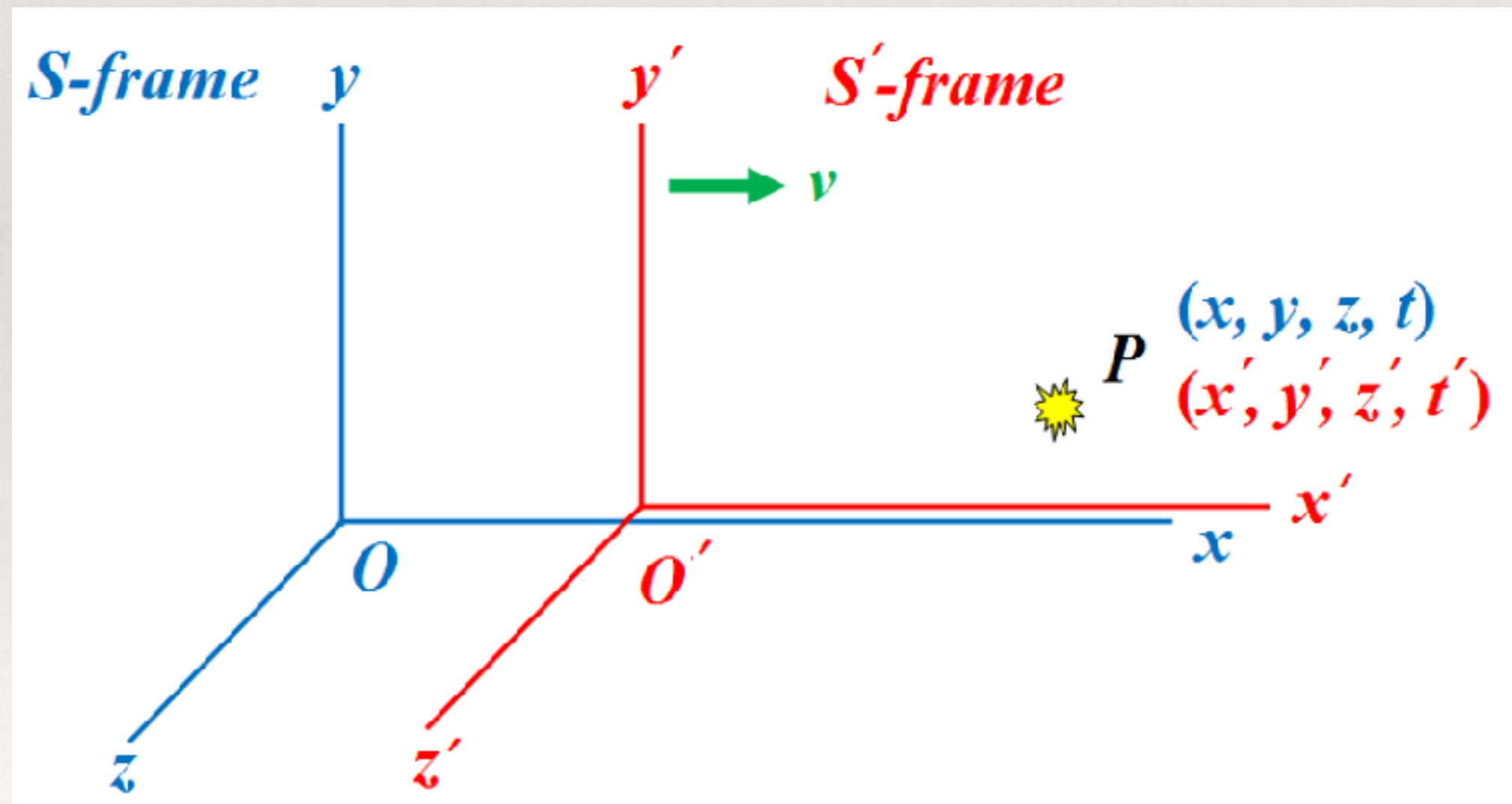
$$x_A - x_B = x'_A - x'_B$$

For simultaneous events

Newton's law and inertial frames

Question: Is Newton's second law consistent with the concept of relativity? More specifically, does Newton's second law remain unchanged under Galilean transformation?

Principle of **Relativity** as stated by Newton:
 "The motion of bodies included in a given space are the same among themselves, whether that space is at rest or moves uniformly forward in a straight line."



$$\frac{dx'}{dt} = \frac{dx'}{dt'}$$

$$\frac{dx'}{dt'} = \frac{dx}{dt} - v$$

$$\frac{dy'}{dt'} = \frac{dy}{dt}$$

$$\frac{dz'}{dt'} = \frac{dz}{dt}$$

$$\frac{d}{dt'}(u_x') = \frac{d}{dt}(u_x - v),$$

$$\frac{du_x'}{dt'} = \frac{du_x}{dt},$$

$$\frac{du_y'}{dt'} = \frac{du_y}{dt},$$

$$\frac{du_z'}{dt'} = \frac{du_z}{dt}.$$

How the velocities are related?

$$u_x' = u_x - v$$

$$u_y' = u_y$$

$$u_z' = u_z.$$

$$\mathbf{u}' = \mathbf{u} - \mathbf{v}.$$

How about acceleration?

$$a_x' = a_x, a_y' = a_y, \text{ and } a_z' = a_z.$$

$$\mathbf{a}' = \mathbf{a}.$$

$$\mathbf{F}' = m\mathbf{a}' = m\mathbf{a} = \mathbf{F}$$

Newton's laws of motion and equation of motion are exactly the same in all 'inertial' systems. The mass **m** is assumed constant.

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t$$

Electromagnetism and relativity

Question : Are the laws of Electromagnetism invariant under Galilean transformation ?

MAXWELL

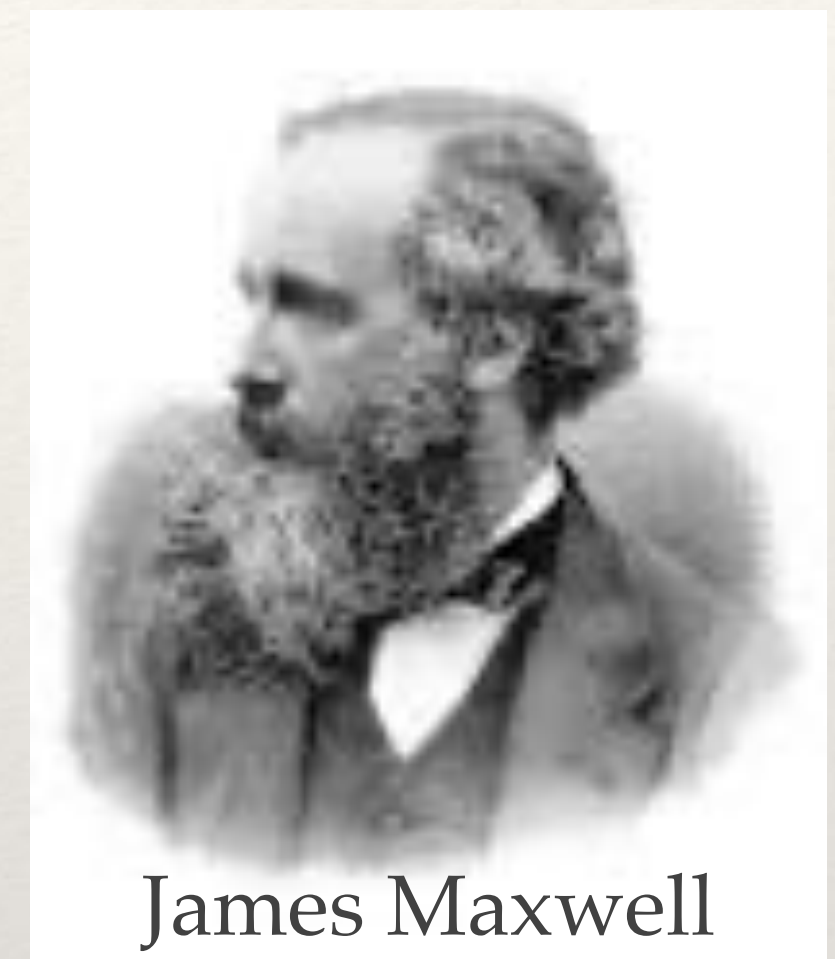
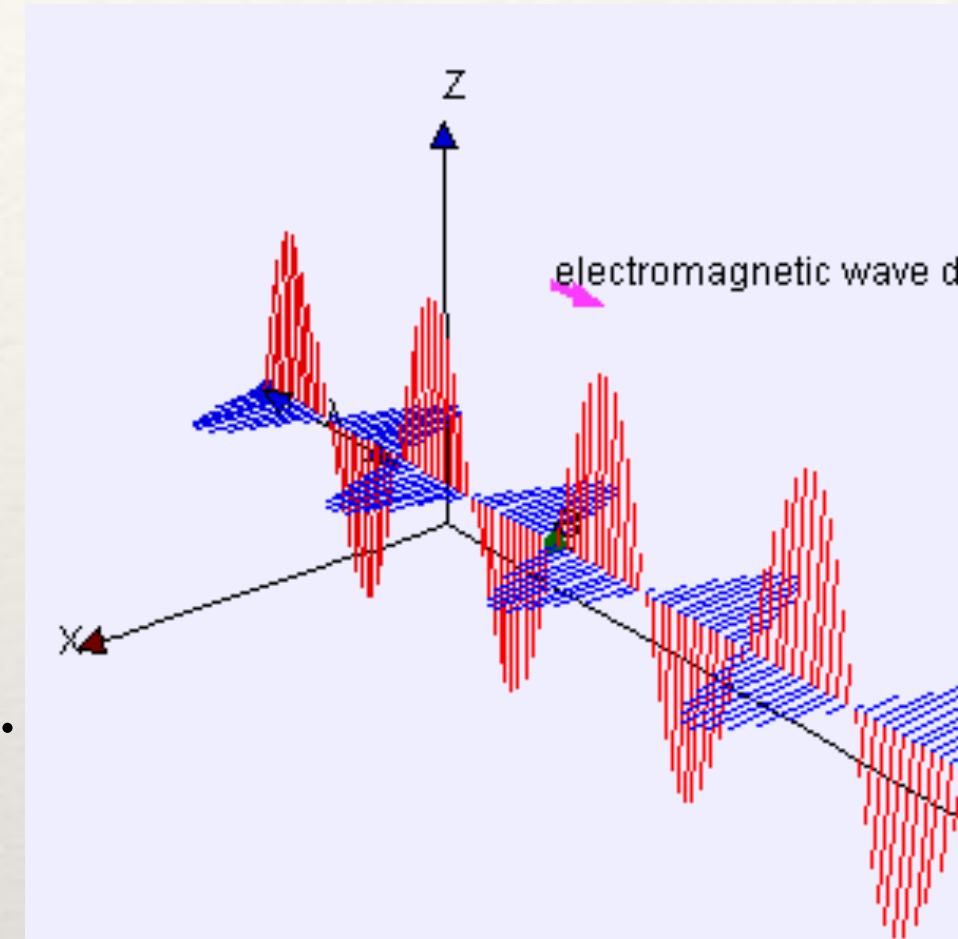
$$\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0,$$
$$\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = 0.$$

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = 0,$$
$$\frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = 0.$$

μ_0 and ϵ_0 are permeability and permittivity of free space.

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ ms}^{-1}.$$

Electromagnetic waves
move with speed of
light in vacuum



James Maxwell

Therefore, a relativity principle exists for mechanics but not for electromagnetism !
We must find the ether frame where laws of electromagnetism is correct.

Consider a second frame S' moving at speed v along the x axis of S , and imagine a beam of light traveling in the same (or opposite) direction. In S the light's speed is $u = c$. Therefore, in S' the speed is

$$u' = c \pm v.$$

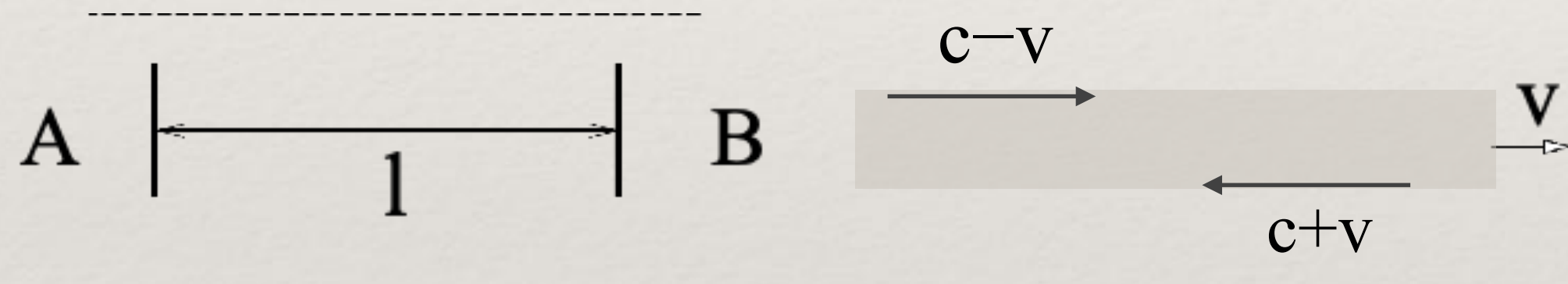
Maxwell's equations do not hold in the inertial frame S' .

A relativity principle exists for both mechanics and electromagnetism, but laws of mechanics as given by Newton needs to be modified.

Resolution: searching for ether winds

- Measure speed of light in different frames.
- Note if there is difference in speed of light in different frame.
- If yes, note if there is a single unique frame where speed is c .

Speed of light is c wrt to ether.



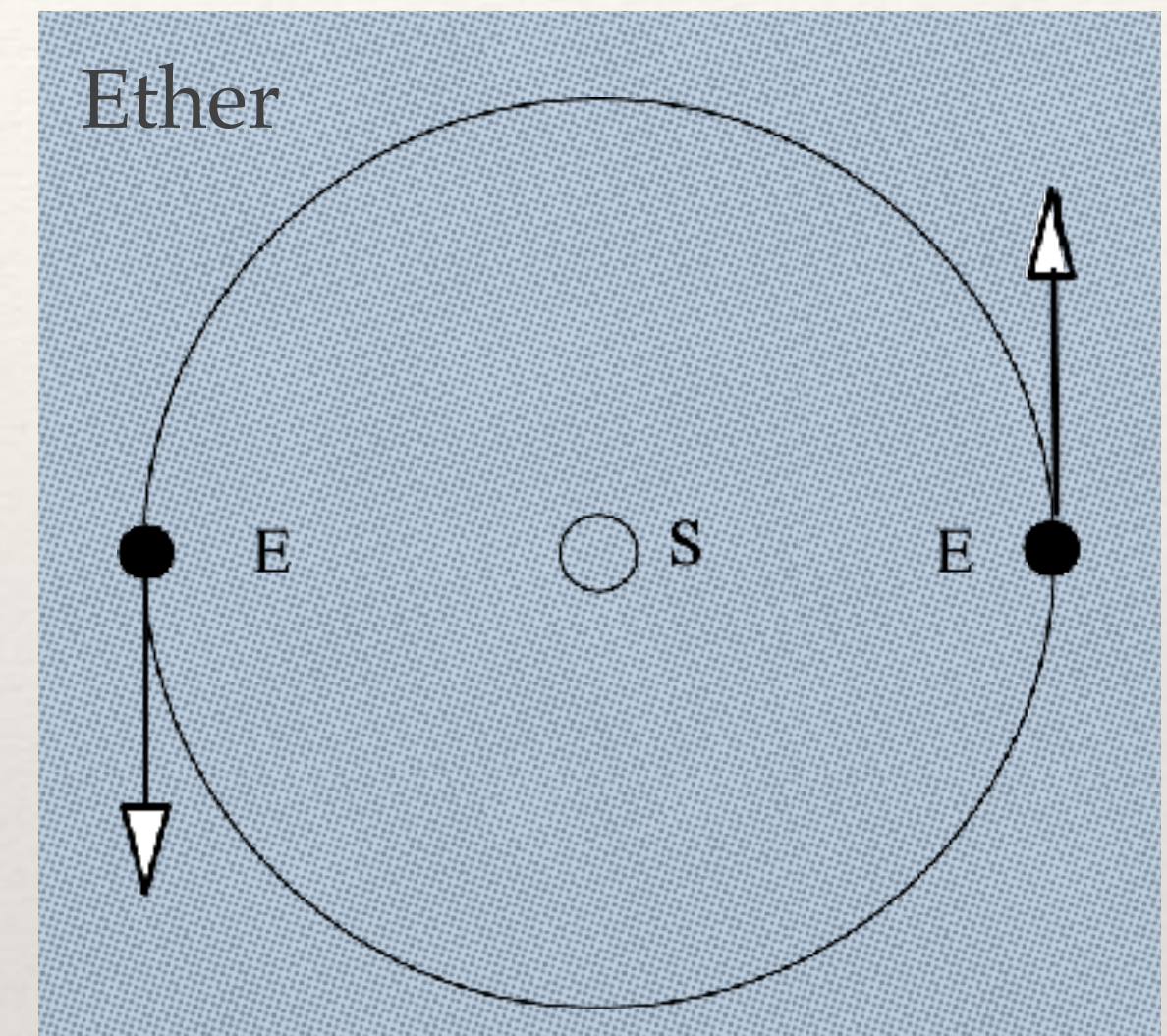
If the apparatus is moving with velocity v , the difference in time required for light to complete a round trip ABA from that of an ether frame will be:

$$\Delta t = \left(\frac{l}{c-v} + \frac{l}{c+v} \right) - \frac{2l}{c}$$

$$= \frac{2l}{c} \frac{v^2}{c^2} \quad (v \ll c)$$

Taylor Expansion: $(1 - v^2/c^2)^{-1} = (1 + v^2/c^2)$

Consider the ether to be fixed with respect to Sun.



Earth moves with 30 km/s reversing the direction every six months.

At any position in the orbit, $\frac{v}{c} = 10^{-4}$

A typical apparatus has, $l = 1 \text{ m}$

Following, $\Delta t = \frac{2l}{c} \frac{v^2}{c^2}$

$$\Delta t = 7 \times 10^{-17} \text{ sec}$$

Insane
precision!

Michelson Morley Experiment (1887)



Albert A. Michelson

Consider first: $l_1 = l_2 = L$

The time for light to go from B to E: t_1

Return time from E to B: t_2

$$ct_1 = L + vt_1, \text{ or } t_1 = L/(c - v)$$

$$ct_2 = L - vt_2, \text{ or } t_2 = L/(c + v)$$

$$\text{Total time: } t_1 + t_2 = t_H = 2 \frac{L/c}{1 - v^2/c^2}$$

See the geometry, C has moved to C'

The time for light to go from B to C: t_3

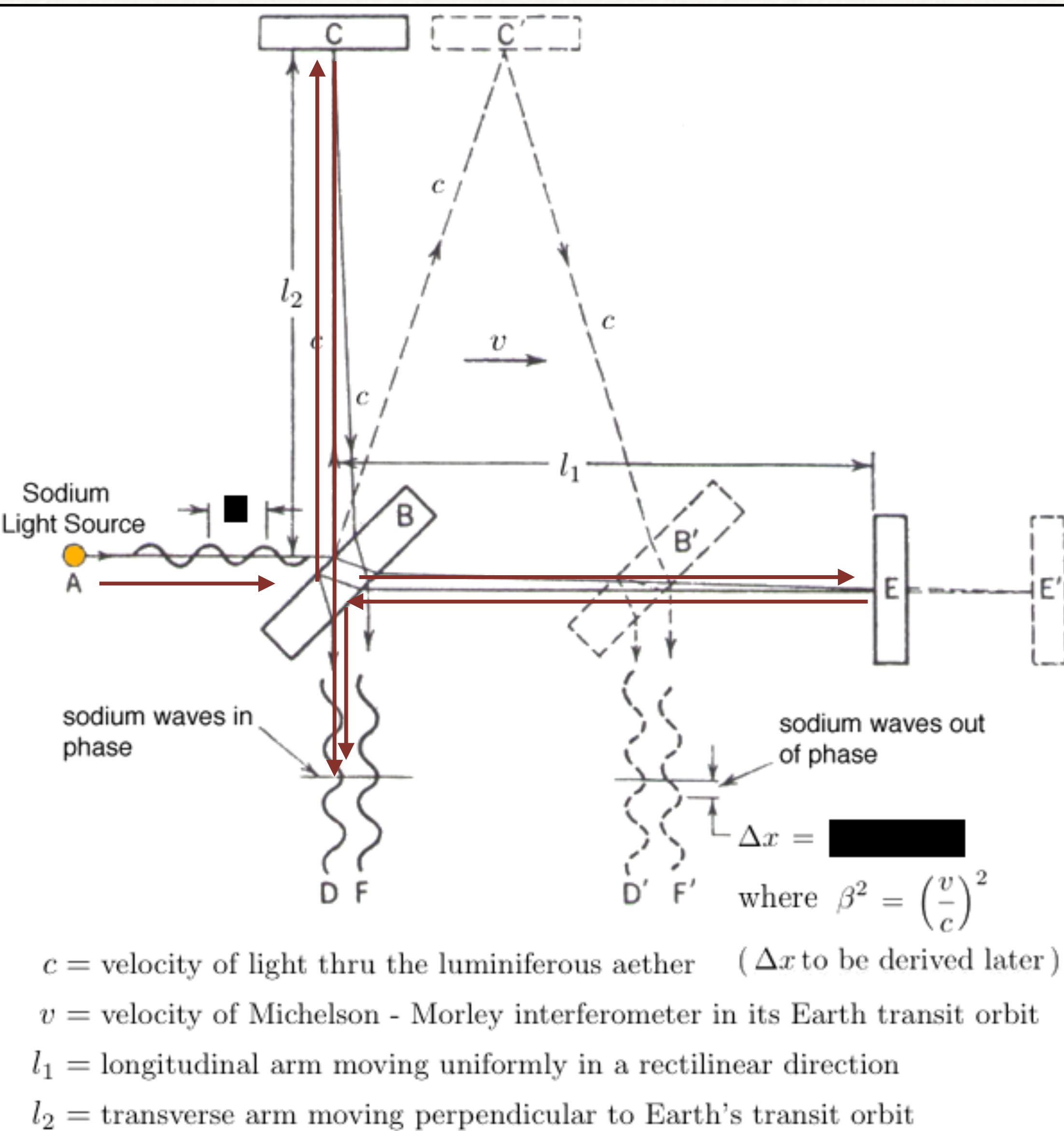
Return time from C to B: t_4

By Symmetry: $t_3 = t_4$

$$(ct_3)^2 = L^2 + (vt_3)^2; t_3 = L/(\sqrt{c^2 - v^2})$$

$$\text{Total time: } t_v = 2t_3 = \frac{2L/c}{\sqrt{1 - v^2/c^2}}$$

All the measurements are with respect to ether frame



Will the emerging beams D and F be in phase or out of phase (Interference)?

Michelson Morley Experiment: Fringe shift

Observations: (A) numerators are identical: represents time taken if the apparatus was at rest
(B) denominators different: (i) represents modifications in time caused by motion of the apparatus, (ii) modifications are not the same.

Practically, $l_1 \neq l_2$

Then turn the apparatus by 90 degrees such that BC is in line of motion and BE is perpendicular to the motion.

There must be a shift in the interference fringes.

H.W: Show that $\Delta N = \frac{\Delta t - \Delta t'}{T} = \frac{l_1 + l_2}{\lambda} \frac{v^2}{c^2}$.

Inference of the Null Result: (i) concept of 'ether' is ruled out
(ii) no such expected change in speed of light.

Nobel Prize in 1917 to Michaelson for the null result!

Observation:

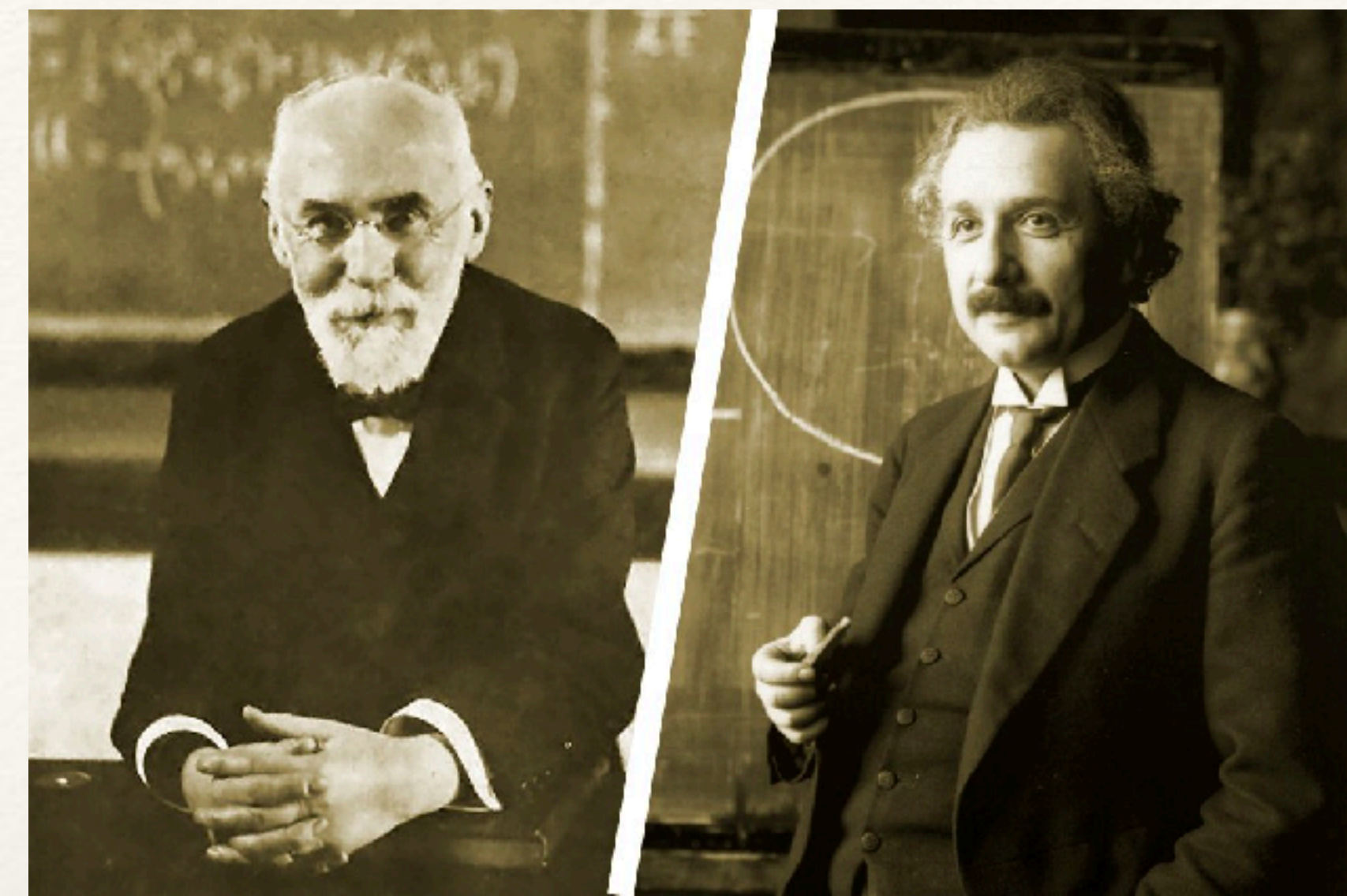
Even though the apparatus was very sensitive, no such fringe shift was found!



Would there be any change in Galilean transformation?

An interesting observation by H. A. Lorentz (In 1892)

And Fitzgerald also proposed the same thing



Suggestion: The length of an object contracts when in motion, along the direction of motion.

Suppose, L_0 is the length of a body at rest.

Then if it moves with speed v parallel to its length, the new length is

$$L_{||} = L_0 \sqrt{1 - v^2/c^2}$$

Contraction of length!

Modification applied to M.M. apparatus: (a) distance B to E is shortened
(b) distance from B to C does not change.

$$t_H = 2 \frac{L_{||}/c}{1 - v^2/c^2} = 2 \frac{L_0/c}{\sqrt{1 - v^2/c^2}} = t_V$$

The time taken are the same !

Explains the null results.....

So far, this contraction hypothesis is just to successfully explain the null result of Michelson Morley experiment.

What is the basis of it ?

Another Observation by Lorentz and Poincare:

Maxwell's equations retain their form under the following set of transformations!

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

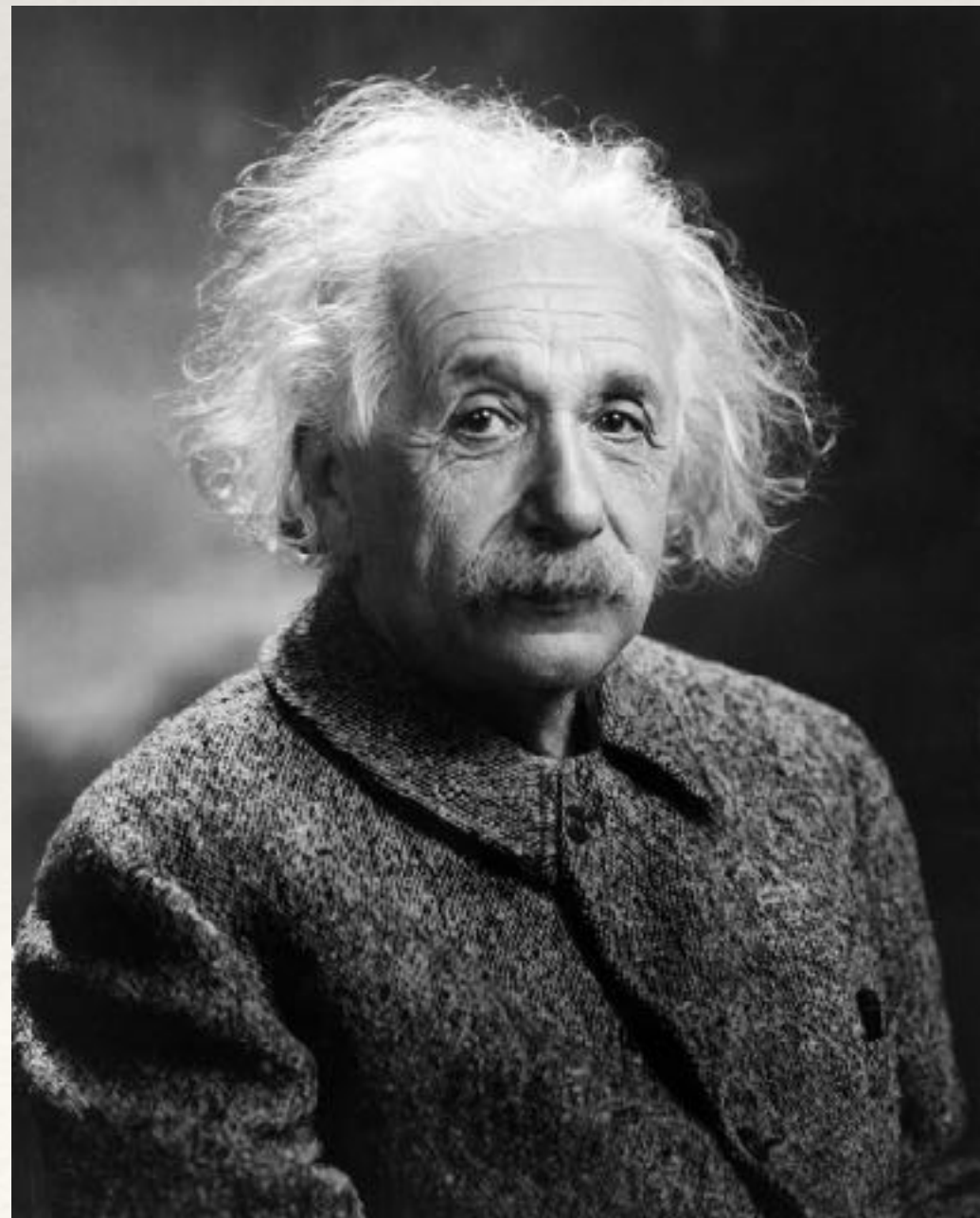
$$t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}.$$

So, what is the fundamental aspect? Need to wait for Einstein (1905).....

The universal velocity and Special Relativity

The findings of Michelson-Morley allows us to say for surly,
``If this ether is real,
Then it has no appeal,
And shows itself off rather poorly.”

Special Theory of Relativity (STR)
satisfactorily answered all the queries.



Postulates of STR

1. The laws of physics are the same in all inertial systems. No preferred inertial system exists. (The Principle of Relativity.)
2. The speed of light in free space has the same value c in all inertial systems. (The Principle of the Constancy of the Speed of Light.)



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Zur Elektrodynamik bewegter Körper

A. Einstein

First published: 1905 | <https://doi.org/10.1002/andp.19053221004> | Citations: 2,021

Summary

- Concept of an Event
- Galilean Transformation (GT) to relate space-time of events in inertial frames
 - Invariance of Newton's laws under GT
 - Non-invariance of Maxwell's equations under GT
 - Michelson Morley Experiment and null result
 - Absence of ether and constancy of velocity of light
- Einstein's postulates of Special Theory of Relativity